

The concept of area; the area of a rectangle

Four properties that every area must have — and the one formula everything else is built on.

LESSON 46

1 × 40 MIN

GEOMETRY 8, ТЕМЫ 45–46

STAGE 9 · 7.1

LESSON CLOCK

5'

12'

12'

7'

4'

Count squares to find an area	5 min
The four properties	12 min
Rectangle and square	12 min
Units and conversions	7 min
Homework	4 min

LEARNING OBJECTIVES

- State the four properties of area.
- Use the formula for the area of a rectangle and of a square.
- Convert between units of area.
- Find a missing side from a given area.

TEXTBOOK

National	Темы 45–46, pp. 100–105
Cambridge	Learner’s Book pp. 142–148
Workbook	Workbook 7.1

01

Explanation

What area is

Area measures how much of the plane a figure covers, in **unit squares**. Whatever definition is used, it must obey four rules:

PROPERTIES OF AREA

1. Area is a **positive** number.
2. **Congruent** figures have **equal** areas.
3. If a figure is cut into parts, the areas of the parts **add up** to the whole.
4. The area of a square of side *1* is *1*.

Property 3 — **additivity** — is the workhorse. It is why a trapezium can be split into a rectangle and two triangles, and why an L-shaped room can be measured as two rectangles.

EQUAL AREAS ≠ CONGRUENT

A 2×8 rectangle and a 4×4 square both have area 16, but you cannot lay one on the other. Rule 2 works in one direction only.

Rectangle, square, and units

$$S_{rectangle} = a \cdot b \mid S_{square} = a^2$$

The formula is really just counting: a 5×3 rectangle holds three rows of five unit squares.

UNIT	IN M ²	USED FOR
1 cm ²	0.0001	a page, a tile
1 m ²	1	a room, a carpet
1 a (are)	100	a garden plot
1 ha	10 000	a field
1 km ²	1 000 000	a district

Lengths scale by k ; areas scale by k^2 . $1\text{ m} = 100\text{ cm}$, so $1\text{ m}^2 = 10\,000\text{ cm}^2$ — a fact that catches out more pupils than any other in this chapter.

02 Worked examples

EXAMPLE
1

A rectangular room is 4.5 m by 3.2 m. Find its area, and the cost of carpet at 85 000 so‘m per m².

1

$S = 4.5 \cdot 3.2 = 14.4\text{ m}^2$

2

$14.4 \cdot 85\,000$

3

$= 1\,224\,000\text{ so‘m}$

ANSWER

14.4 m², 1 224 000 so‘m

EXAMPLE 2 A rectangle has area 84 cm^2 and one side 7 cm . Find its perimeter.

① $84 \div 7 = 12$
The other side.

② $P = 2(7 + 12)$

③ $= 38 \text{ cm}$

ANSWER

38 cm

EXAMPLE 3 An L-shaped floor is a 6×4 rectangle with a 2×2 square cut from one corner. Find its area.

① $6 \cdot 4 = 24$
The whole rectangle.

② $2 \cdot 2 = 4$
The corner removed.

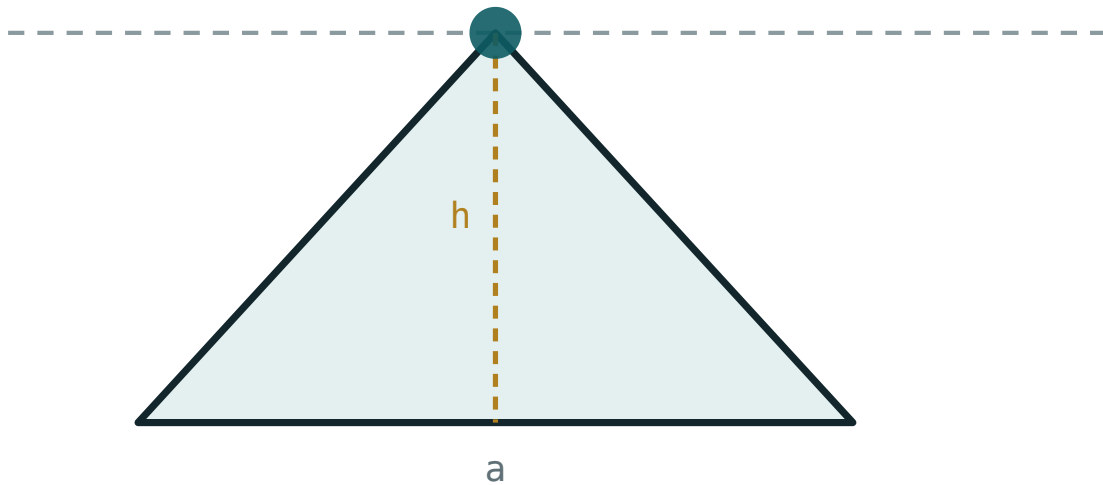
③ $24 - 4 = 20$
Additivity, used backwards.

ANSWER

20 m^2

03 Interactive model

Change the base and height and watch the area follow the product.

Base times heightbase a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6****04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Area	Yuza	Площадь
Unit square	Birlik kvadrat	Единичный квадрат
Congruent figures	Teng figuralar	Равные фигуры
Additivity of area	Yuzaning additivligi	Аддитивность площади
Rectangle	To'g'ri to'rtburchak	Прямоугольник
Square	Kvadrat	Квадрат
Square metre	Kvadrat metr	Квадратный метр
Hectare	Gektar	Гектар
Dimensions	O'lchamlar	Размеры

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The area of a 7×9 rectangle is:

1 m^2 equals:

A square of area 49 cm^2 has perimeter:

Congruent figures:

may have different areas

always have equal areas

always are rectangles

have equal perimeters but not areas

1 ha equals:

100 m²

1 000 m²

10 000 m²

100 000 m²

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find the area of a 5 cm by 8 cm rectangle.

ANSWER 40 cm²

2. Find the area of a square of side 9 cm.

ANSWER 81 cm²

3. Convert 3 m² to cm².

ANSWER 30 000 cm²

4. A rectangle has area 36 and one side 4. Find the other.

ANSWER 9

5. Find the area of a 2.5 m by 4 m rectangle.

ANSWER 10 m²

6. Convert 2 ha to m².

ANSWER 20 000 m²

7. Find the side of a square of area 144 cm².

ANSWER 12 cm

Medium · the standard question

7 PROBLEMS

1. A room 5.5 m by 4 m is carpeted at 90 000 so‘m/m². Find the cost.

ANSWER 1 980 000 so‘m

2. A rectangle has area 150 cm² and perimeter 50 cm. Find its sides.

ANSWER 10 cm and 15 cm

3. An L-shape is a 8×5 rectangle less a 3×2 corner. Find its area.

ANSWER 34

4. Convert 45 000 cm² to m².

ANSWER 4.5 m²

5. A square field has area 1 ha. Find its side.

ANSWER 100 m

6. A rectangle 12 cm by 5 cm is cut into two equal squares plus a rectangle. Is that possible? Explain.

ANSWER Two 5×5 squares use 10 cm of the length, leaving a 2×5 rectangle — yes.

7. The sides of a rectangle are doubled. What happens to the area?

ANSWER It becomes four times as large.

Hard · stretch and reason

7 PROBLEMS

1. A rectangle has perimeter 40 cm. Find the dimensions giving the greatest area.

ANSWER 10×10 — the square, area 100 cm^2

2. A path 1 m wide runs round the outside of a 10 m by 6 m garden. Find the area of the path.

ANSWER $12 \cdot 8 - 10 \cdot 6 = 36 \text{ m}^2$

3. A rectangle has area 60 cm^2 and its length is 4 cm more than its width. Find its sides.

ANSWER 6 cm by 10 cm

4. A field of 2.4 ha is rectangular with one side 120 m. Find the other side and the perimeter.

ANSWER 200 m, perimeter 640 m

5. A floor 4.2 m by 3.6 m is tiled with 30 cm squares. How many tiles are needed?

ANSWER $14 \cdot 12 = 168$

6. The area of a square is numerically equal to its perimeter. Find its side.

ANSWER $a^2 = 4a$, so $a = 4$

7. Explain why two figures with equal areas need not be congruent.

ANSWER Area is a single number and loses all information about shape; a 1×12 rectangle and a 3×4 rectangle share an area but no dimension.

07 Homework

Homework — 5 problems

Geometry 8, Темы 45–46, pp. 100–105. Always write the unit.

- Find the area of a 7.5 m by 6 m room.
- Convert 5 m^2 to cm^2 and $30\,000 \text{ cm}^2$ to m^2 .
- A rectangle has area 96 cm^2 and one side 8 cm. Find its perimeter.
- An L-shape is a 10×7 rectangle less a 4×3 corner. Find its area.

5. A square has area 225 m^2 . Find its side and perimeter.